High Frequency Response of Non-Volatile Memristors

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***Abstract*— This paper presents an analytical investigation of the transient and steady-state response of non-volatile memristors to high frequency periodic inputs, using as a case study a TaO_X-based nano-scale memristor model derived at HP Labs. For the first time, we provide a mathematical proof for the fading memory phenomenon in memristors stimulated by periodic inputs in the high frequency limit. Specifically, we demonstrate that the steady-state response of a non-volatile memristor, exhibiting asymmetric switching kinetics with respect to the polarity of the input, depends only on the amplitude of the testing signal and not on the device initial conditions. Based on the results of our analyses, we provide an alternative method for tuning the memristor state by using high-frequency AC inputs, and introduce a new system-theoretic visualization tool, namely the input-referred High-Frequency Dynamic Route Map (HF-DRM), that allows the reproduction of the memristor time-response to any high-frequency periodic input from each admissible initial condition. The purely theoretical results introduced in this paper could inspire new approaches for modulating the memory states of practical non-volatile memristors.**

***Index Terms*— Memristor, model, non-volatile, high frequency, analytical equations, HF-DRM, steady-state response, transient response.**

I. Introduction

INDISPUTABLY, circuit and system theories have laid the foundations for the remarkable achievements of modern electronics. Simply stated, it would be impossible to design integrated circuits or test current nano-electronic systems by merely fiddling and tweaking their parameters, as these consist of billion of devices that very often exhibit complex dynamical behaviors. Employing system-theoretic tools may allow to gain a deeper understanding of the highly-nonlinear dynamics of fabricated memristive devices, and to develop efficient analysis tools, employable in standard simulation environments, enabling to harness their peculiar behaviors for circuit design. Looking to the future, circuit theory will once more play a key role in the realization and development of cutting-edge technologies such as artificial intelligence, cognitive science and robotics, to name just a few examples. These fields heavily rely on the real-time processing and interpretation of huge banks of data in the form of electrical signals, which is a cumbersome task that could be greatly facilitated by the adoption of emerging electronic device technologies, such as ReRAMs, MRAMs, Ferro-Electric Memories, Phase Change Memories, Atomic Switches etc. Ultimately, the design of circuits, leveraging the capabilities of these devices, requires a set of model based mathematical formulas, that can be used to predict all the possible operating regimes [1], [2]. Fortunately, the circuit-theoretic introduction of the memristor as the fourth fundamental two-terminal electrical circuit element [3], [4] provided a unifying mathematical framework, allowing to group all the aforementioned disruptive resistance switching memories into appropriate classes [5], which may enable the development of proper circuit models.

The memristor is a two-terminal electronic device exhibiting a *pinched hysteresis loop*, that is a hysteretic current-voltage locus that passes through the origin, under purely-AC periodic current or voltage stimulation. Furthermore, exceeding a certain frequency, the lobes' areas of the pinched hysteresis loop decrease monotonically as the input frequency increases. The memristor current-voltage characteristic, induced by inputs of this kind, reduces to a single-valued curve [6], when the input frequency is sufficiently high. Based on Chua's theoretical framework, the form of the current-voltage curve in the high frequency limit, identifies the high-level mathematical model that best describes the memristor under study. Specifically, a physical memristor displaying a current-voltage locus that illustrates a straight line (single valued function), passing through the origin, under a purely-AC high-frequency current or voltage input, may be classified as a *generic (extended)* memristor [7]. In the simplest first-order case, the fundamental circuit-theoretic properties of a *generic (extended)* memristor, can be captured by the following differential algebraic equation (DAE) set:

$$i = G(x)\upsilon \;\; (i = G(x,\upsilon)\upsilon), \tag{1}$$

$$\frac{dx}{dt} = g(x,\upsilon). \tag{2}$$

The variable i denotes the current flowing through the device, υ is the voltage across its terminals, while x corresponds to its state. The functions $G(\cdot)$ and $g(\cdot)$ are referred to as memductance and state evolution function, respectively. Very interestingly, investigations, performed on several physical memristor models, showed that the slope (shape) of the non-hysteretic steady-state current-voltage linear (single-valued

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nonlinear) locus of a *generic (extended)* memristor in the high frequency limit depends only on the amplitude of the testing signal [7], [8]. In other words, the steady-state behavior of the device, under the aforementioned stimulation scenario, is unique, for a given amplitude of the input signal, irrespective of its initial condition. In fact, this input-induced memory loss phenomenon [9], called history erase effect,[1] was first discovered in memristors in [11], on the basis of experimental data extracted from a TaO_x-based nanoscale memristor, manufactured at Hewlett Packard (HP) Labs, as well of an in-depth circuit- and system-theoretic analysis of its model. The frequency of each testing stimulus, in the aforementioned study, was set so low as to allow the device to adapt its behavior to the periodic excitation, as revealed by the appearance of a hysteretic locus on the current-voltage plane in a scenario of this kind. However, predicting the behavior of an intrinsically-complicated memristive dynamical system, under a given periodic input, is a particularly challenging task, which explains why no mathematical proof for the appearance of the history-erase phenomenon in physical memristors has been provided in the literature to date. Here, we demonstrate that the study of the transient and steady-state dynamics of memristors becomes tractable when the memristive response is induced by high-frequency periodic inputs. Consequently, we derive an analytical formula that captures mathematically the condition for the emergence of fading memory in memristors under such stimuli. Furthermore, we disclose a hidden property of the system-theoretic Dynamic Route Map (DRM) analysis tool [12], which was recently hinted at, in [13]. Specifically, its capability to predict the mean value of the steady-state oscillation in the memory state of a given nonvolatile memristor under any purely-AC periodic voltage (current) stimulus, featuring any shape (e.g. triangular, sinusoidal, or rectangular), as it falls across (flows through) its physical stack. Inspired by the DRM, we then introduce a new graphical tool, namely the input-referred *High-Frequency Dynamic Route Map (HF-DRM)*, which allows to determine important characteristics of the steady-state oscillation in the state variable, namely its mean and peak-to-peak values, over each cycle of a high-frequency periodic signal, which induces fading memory effects in the memristor under focus. The study, reported in this manuscript, provides general valuable insights on the high-frequency response of typical non-volatile memristors, despite here a representative nanodevice from this class is chosen as the object of the investigations [14]. The mathematical model of this nanodevice is briefly presented in Section II, which also discusses important dynamical behaviors featured by non-volatile sign-invariant memristors under high-frequency periodic inputs, including the fading memory phenomenon. Based on the behavioral observations highlighted in Section II, Section III outlines the main findings of our research deriving approximate analytical equations that respectively capture the steady-state (see III-A) and transient (see III-B) response of the non-volatile memristor under high-frequency periodic inputs. The formulas, proposed in Section III, are also validated under several stimulation scenarios, specifically for purely-AC periodic square-wave, sinusoidal, and triangular voltage inputs, as well as for AC periodic square-wave voltage inputs that include various DC offsets. Especially with regard to the steady-state high-frequency response of the TaO_x memristor, in view of a future potential application, we exploit the fading memory property to develop a methodology by which the device can be configured into a desired memory state, on the basis of the aforementioned analytical equations. FThe HF-DRM visualization tool mentioned earlier, is employed throughout this section to gain a deep understanding of the high frequency input-induced fading memory phenomenon, in view of the aforementioned future potential application. Section IV drafts discussions and conclusions.

[1] In nonlinear system theory, the history erase effect phenomenon is known as fading memory [10].

II. Memristive Fingerprints and Fading Memory in a TaO_x Memristor From HP Labs

According to the Strachan model [14], the differential algebraic equation (DAE) set of the TaO_x memristor under study is expressed as

$$i = G(x, \upsilon)\upsilon = \left(G_m \cdot x + a \cdot e^{b\sqrt{|\upsilon|}}(1-x)\right)\upsilon, \qquad (3)$$

$$\frac{dx}{dt} = g(x, \upsilon) = \begin{cases} g^{(+)}(x, \upsilon) & \text{for } \upsilon \geq 0 \\ g^{(-)}(x, \upsilon) & \text{for } \upsilon < 0 \end{cases} \qquad (4)$$

where

$$g^{(+)}(x, \upsilon) = B\sinh\left(\frac{\upsilon}{\sigma_{on}}\right) e^{-\frac{x^2}{x_{on}^2}} e^{\frac{G(x,\upsilon)\upsilon^2}{\sigma_p}}, \qquad (5)$$

and

$$g^{(-)}(x, \upsilon) = A\sinh\left(\frac{\upsilon}{\sigma_{off}}\right) e^{-\frac{x_{off}^2}{x^2}} e^{\frac{1}{1+\beta G(x,\upsilon)\upsilon^2}}, \qquad (6)$$

in which x is a unit-less state variable constrained to lie within the closed range $x \in [0, 1]$ at all times. The state evolution function $g(\cdot)$ in this device is sign-invariant, i.e. $g(x, \upsilon) > (<) 0 \ \forall \ \upsilon > (<) 0$. It is important to note that the analyses presented throughout this manuscript may be applied on any other real-world non-volatile memristor model exhibiting a sign-invariant state evolution function [15], [16], [17], [18], etc. Fig. 1(a) illustrates a cross section of the physical device characterized and modeled by *Strachan et al.* in [14]. The resistive state of the device is typically configured by applying a suitable voltage pulse across the critical TaO_x channel through the Pt and Ta electrodes. The paper, where the model was originally proposed, provides a detailed description of the device operating principles, as well as the physical interpretation of equations (3)-(6). The model parameter values are listed in Table I.[2]

A. Memristive Fingerprints

We simulated the model DAE set, expressed by equations (3)-(6), by applying a zero-mean sinusoidal voltage input of the form $\upsilon = \hat{\upsilon} \cdot \sin(2\pi f t)$ (see the inset in Fig. 1(b)) with amplitude $\hat{\upsilon} = 0.5\,\text{V}$, each frequency f within the set $\{3 \cdot 10^4, 15 \cdot 10^4, 5 \cdot 10^5, 10^8\}$ Hz, and initial condition $x(0) \triangleq$

[2] The Strachan TaO_x memristor model may experience numerical stability issues, which have been recently resolved in [19] through an appropriate reformulation of the original equations, while preserving their accuracy.

TABLE I
PARAMETER VALUES OF THE TaO_X MEMRISTOR MODEL FROM STRACHAN ET AL

A/s^{-1}	σ_{off}/V	x_{off}	$\beta/(\mathrm{A}^{-1}\mathrm{V}^{-1})$
10^{-10}	0.013	0.4	500
B/s^{-1}	σ_{on}/V		x_{on}
10^{-4}	0.45		0.06
$\sigma_p/(\mathrm{AV})$	G_m/S	a/S	$b/\mathrm{V}^{-1/2}$
$4 \cdot 10^{-5}$	0.025	$7.2 \cdot 10^{-6}$	4.7

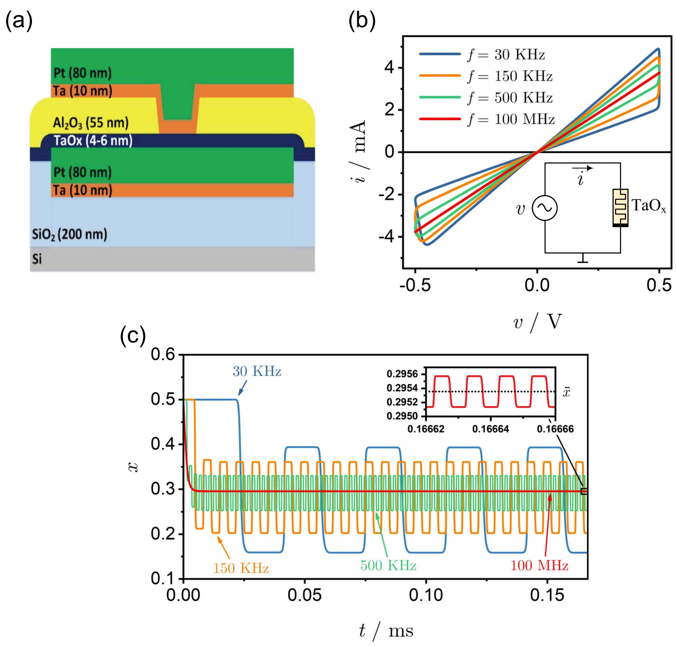

Fig. 1. (a) Cross section of the TaO_x-based memristor from HP Labs. This figure was reprinted with permission from [11]. (b) and (c) Dynamics of the TaO_x nanodevice under a sinusoidal voltage input of the form $v = \hat{v}\sin(2\pi f t)$ in the test circuit shown in the lower right corner of panel (b). Panel (b) specifically shows steady-state Lissajous $i - v$ curves obtained for $\hat{v} = 0.5\,\mathrm{V}$ and $f \in \{3 \cdot 10^4, 15 \cdot 10^4, 5 \cdot 10^5, 10^8\}\,\mathrm{Hz}$. In all cases, the memristor state variable x was initialized at $x_0 = 0.5$. Panel (c) illustrates the time waveform of the device state for each of the periodic input signals in (b). The inset panel in (c) zooms in across the last four periods of the state response for the highest input frequency employed. For this input frequency, at steady-state, the memristor state oscillates with a very small peak-to-peak amplitude about the mean value $\bar{x}_s = 0.29535$ (see the dotted horizontal black line in the inset of panel (c)).

x_0 equal to 0.5, across the device terminals. Fig. 1(b) illustrates the resulting steady-state current-voltage plots. Notably, the hysteresis lobe area decreases monotonically as the excitation frequency increases, and for $f = 100\,\mathrm{MHz}$ (see the red-colored locus) the current-voltage plot appears as a single-valued function through the origin. Panel (c) captures the time-waveforms of the state variable $x(t)$ for each of the aforementioned simulation settings. Starting from the chosen initial condition, all the state responses exhibit transitory dynamics, converging asymptotically to steady-state periodic oscillations. In agreement with the $i - v$ plot frequency modulation, inferable from panel (b), the steady-state oscillation amplitude of the state reduces with increasing frequency in panel (c). According to the asymptotic state response, shown in the inset of panel (c), for $f = 100\,\mathrm{MHz}$, x oscillates with a very small peak-to-peak amplitude (<0.0005) about the mean value $\bar{x}_s = 0.29535$, which is why the corresponding red-colored current-voltage plot in panel (b) exhibits no apparent hysteresis [7]. Under these circumstances we can therefore assume that $x(t)$ *is practically equal to* $\bar{x}_s$ *for all times, after transients decay to zero*.

B. Fading Memory

Physical device experiments [11] as well as simulations [20] have already shown that the TaO_x nanodevice from HP Labs exhibits a fading memory under suitable periodic excitation. Here, we verify this device feature under high frequency AC inputs. But before proceeding, let us first establish a definition for what we consider to be a *high-frequency* periodic input throughout this manuscript:

A high-frequency periodic input induces a solution waveform of the device state, which, after some finite time, becomes periodic with such a small peak-to-peak amplitude that the resulting device current-voltage locus exhibits no apparent hysteresis.

At this point, it is important to clarify that the specification of the input frequency value, on its own, is not sufficient to characterize a stimulus as a high-frequency periodic input. The amplitude of the AC input plays a crucial role as well. This observation will be further analyzed a bit later in this section (see Fig. 3(b)). The colored x versus t loci in Fig. 2(a), were numerically derived by applying a periodic signal of the form $v = (0.5\,\mathrm{V})\sin\left(2\pi(10^8\,\mathrm{Hz})t\right)$ directly across the nanodevice, and integrating the model DAE set for each initial condition x_0 in the set $\{0.2, 0.4, 0.6, 0.8, 1\}$. Each of the colored state responses become periodic at the same frequency as the input at $t = nT = n/f$ with $x(nT) = 0.29513$, where n is a positive integer that takes distinct values depending on the initial condition. For our case study, n is equal to 1758, 1228, 1390, 1514, and 1629 when x_0 is initialized at 0.2, 0.4, 0.6, 0.8, and 1, respectively (see Fig. 2(b)). As illustrated by the black-shaded response in Fig. 2(a), when the initial condition x_0 for the memory state is equal to 0.29513, the solution waveform $x(t)$ is periodic at all times [8]. This is more clearly captured in the upper panel of Fig. 2(c), which zooms in across the first two periods of this state response. The lower panel in Fig. 2(c) plots the input voltage waveform over the same time interval (see the blue curve). Obviously, all the state waveforms converge at $t = 1758T$, that is once the state response for $x_0 = 0.2$ becomes periodic. At this time instant, all the initial conditions, assigned to the memory state, are erased (see panels (a) and (d) in Fig. 2 and the description in the caption of the same figure for further details).

III. STEADY-STATE AND TRANSIENT RESPONSE OF NON-VOLATILE SIGN-INVARIANT MEMRISTORS TO HIGH FREQUENCY PERIODIC INPUTS

Here, we exploit the insights gained through the simulations, performed in the preceding section, to derive analytical equations that describe the steady-state, as well as the

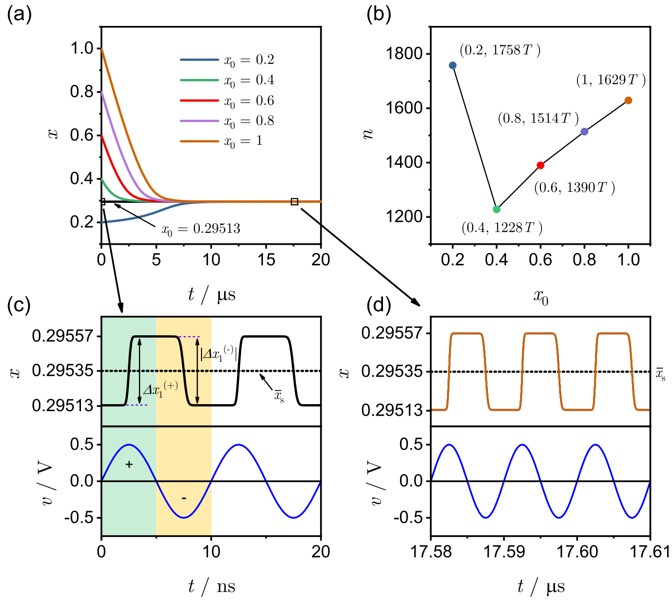

Fig. 2. Fading memory effects emerging in the TaO_x-based memristor from HP Labs under a sinusoidal voltage input $v = \hat{v}\sin(2\pi f t)$ with $\hat{v} = 0.5$ V and $f = 100$ MHz. Panel (a) shows the memristor state responses for a set of initial conditions, specifically $x_0 \in \{0.2, 0.29513, 0.4, 0.6, 0.8, 1\}$. The n versus x_0 plot in panel (b) indicates the number n of input cycles necessary for the state response in (a) to become periodic, for each initial condition from the aforementioned set, except the case $x_0 = 0.29513$. The black-shaded locus in (c) zooms in across the first two periods of the state waveform initiated from $x_0 = 0.29513$ in (a). The black double headed arrow within the green (yellow) region in panel (c), indicates the change in the memristor state $\Delta x_1^{(+)}$ ($\Delta x_1^{(-)}$) over the first positive (negative) half period of the input voltage waveform, which is shown as a blue-colored locus below the x versus t response in (c). The upper plot in (d) provides a detailed view of any of the state time waveforms from (a) within the time interval $[1758T, 1761T]$. At $t = 1758T$, all the waveforms converge to the same periodic oscillation. The blue curve in the lower part of panel (d) shows the input voltage waveform within the same time interval considered for the state response in the upper part of the same panel.

transient response of non-volatile sign-invariant memristors to high-frequency periodic inputs.

A. Steady-State Response

Assuming that the device is stimulated by an AC periodic voltage applied across its terminals, the change in x over the n^{th} input cycle can be expressed as

$$\Delta x_n = \Delta x_n^{(+)} + \Delta x_n^{(-)}, \tag{7}$$

where

$$\Delta x_n^{(+)} = \int_{(n-1)T}^{(n-1)T+T/2} g^{(+)}\big(x(t), v(t)\big)\, dt, \tag{8}$$

and[3]

$$\Delta x_n^{(-)} = \int_{(n-1)T+T/2}^{(n-1)T+T} g^{(-)}\big(x(t), v(t)\big)\, dt, \tag{9}$$

where n is a positive integer. $\Delta x_n^{(+)}$ and $\Delta x_n^{(-)}$ represent the change in memristor state over the positive and negative input half period within the n^{th} cycle, respectively. By definition, when $x(t)$ is periodic about a mean value $\bar{x}_s$, the net change of the memristor state (and thus also of its memductance) per input cycle is zero, and the relationship between $\Delta x_n^{(+)}$ and $\Delta x_n^{(-)}$ reduces to

$$\Delta x_n^{(+)} + \Delta x_n^{(-)} = 0. \tag{10}$$

As demonstrated by the red-colored time response of the memristor state variable in Fig. 2(d), at steady-state, x may oscillate with a negligible peak-to-peak amplitude, about a certain mean value $\bar{x}_s$, under an AC periodic input of sufficiently-high frequency [8]. In this case, $\Delta x_n^{(+)}$ and $\Delta x_n^{(-)}$ can be calculated very accurately by changing $x(t)$ with the respective mean value $\bar{x}_s$ in equations (8) and (9). Furthermore, assuming that the initial condition x_0 for the memory state is chosen in such a way that the waveform of $x(t)$ is periodic at all times, without loss of generality, the integrals in (8) and (9) can be calculated over the first period of the zero-mean AC periodic input by setting $n = 1$ (see the colored regions in Fig. 2(c)). Therefore, under these assumptions, combining (8), (9), and (10), and removing the subscript identifier n (here equal to 1) to avoid clutter, yields:

$$\Delta x^{(+)} + \Delta x^{(-)} = 0, \tag{11}$$

where

$$\Delta x^{(+)} = \int_0^{T/2} g^{(+)}\big(\bar{x}_s, v(t)\big)\, dt, \tag{12}$$

and

$$\Delta x^{(-)} = \int_{T/2}^{T} g^{(-)}\big(\bar{x}_s, v(t)\big)\, dt. \tag{13}$$

As it will be shown shortly, (11)-(13) allow to calculate precisely the mean value $\bar{x}_s$ of the oscillating waveform of the state variable at steady-state, for a given amplitude $\hat{v}$ of a high-frequency zero-mean periodic input. Most importantly, the mere use of this set of equations to infer the level $\bar{x}_s$, around which the memory state is found to oscillate asymptotically, will clearly demonstrate that, for memristive systems exhibiting a fading memory under high-frequency periodic excitations, $\bar{x}_s$ *is independent of the initial condition* x_0 *on the memory state itself.*

Let us now validate the predictive capability of equations (11)-(13) under different high-frequency periodic excitation scenarios. The memristor model under test will be, once more, the TaO_x-based memory device from HP Labs.

1) Square-Wave Purely-AC Periodic Voltage Inputs: Investigating the response of a non-volatile memristor under some excitation scheme, that involves voltage pulses, is particularly interesting, as such signals are broadly used for electroforming, characterizing, and, very importantly, tuning the conduction state of a device of this kind in real-world experiments. We will therefore start testing the predictive power of equations (11)-(13) by assuming that the input corresponds to a high-frequency square-wave periodic voltage signal, alternating between two voltage levels $+V$ and $-V$ of opposite

[3]Throughout this paper the input voltage signal applied across the memristor device, is assumed to undergo a positive (negative) excursion over the first (second) half period in each cycle. Similar conclusions apply *mutatis mutandis* for inputs shifted by 180° degrees as compared to the ones considered in this manuscript.

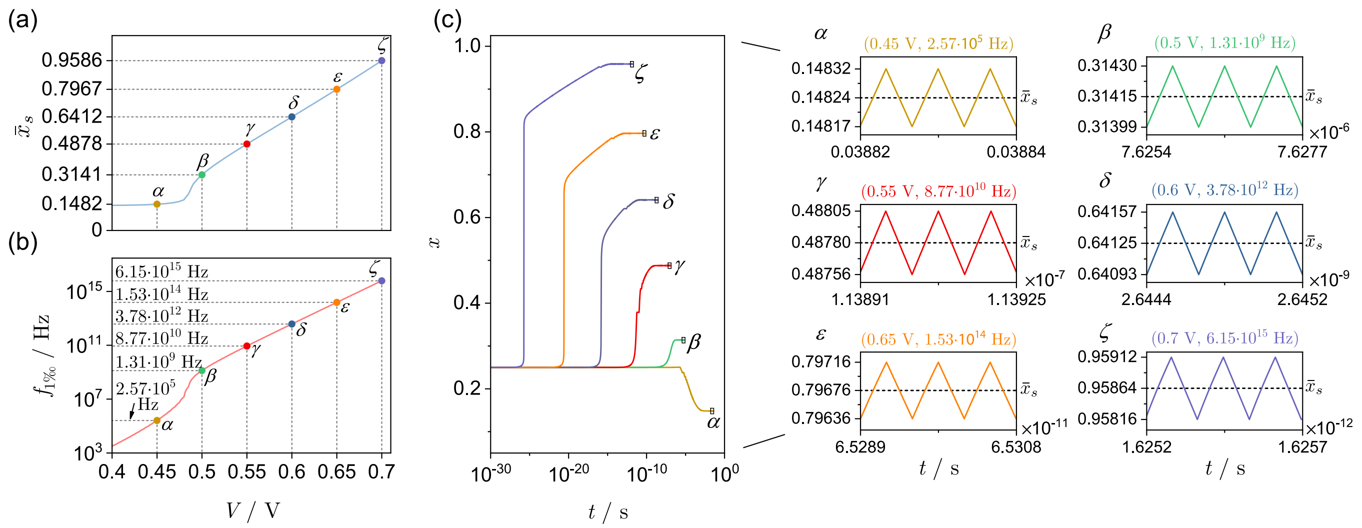

Fig. 3. Validation of equations (14)-(16) under the assumption that high-frequency square-wave purely-AC periodic voltage inputs are applied between the TaO_x-based memristor terminals. The light blue-colored curve in panel (a) captures the mean value $\bar{x}_s$ of the steady-state oscillation of the memristor state as a function of the square-wave input amplitude V, swept across the closed range [0.4, 0.7] V. The light red-colored locus in panel (b) illustrates the frequency $f_{1‰}$ as a function of the input amplitude V, tuned across the same closed range as in (a) (a definition for $f_{1‰}$ is provided in the text body). Each colored round symbol in panels (a) and (b) (see, respectively, points α, β, γ, δ, ϵ, and ζ) indicates the mean value $\bar{x}_s$ and the frequency $f_{1‰}$ corresponding to a square-wave stimulus of one of the amplitudes in a set of six levels (particularly, in turn, 0.45 V, 0.5 V, 0.55 V, 0.6 V, 0.65 V, and 0.7 V). The six x versus t loci, illustrated in panel (c), were obtained by integrating the memristor DAE set under six square-wave voltage inputs featuring the amplitude-frequency pairs corresponding to the colored round symbols in panel (b). In all the simulations the memristor state was initialized at $x_0 = 0.25$. The inset figures $\alpha - \zeta$ in panel (c) zoom in across the last three periods of the model state solutions from the same panel. The horizontal dashed line in each of these insets indicates the mean value of the corresponding oscillation in the device state at steady state.

polarity, over each cycle of period T. Under this hypothesis, calculating the integrals, equations (12) and (13) reduce to

$$\Delta x^{(+)} = g^{(+)}(\bar{x}_s, V) \cdot \frac{T}{2}, \tag{14}$$

and

$$\Delta x^{(-)} = g^{(-)}(\bar{x}_s, -V) \cdot \frac{T}{2}, \tag{15}$$

respectively. Substituting (14) and (15) into (11) yields

$$g^{(+)}(\bar{x}_s, V) + g^{(-)}(\bar{x}_s, -V) = 0. \tag{16}$$

Notably, (16) links the mean value $\bar{x}_s$ of the steady-state oscillation in the state variable x to the amplitude V of the zero-mean high-frequency square-wave periodic input, being independent of the input frequency $f = 1/T$ as well as of the initial condition x_0. Regardless, equation (16) is valid if and only if the input can be characterized as a *high-frequency* periodic signal according to the definition provided earlier in this section. Fig. 3(a) shows the $\bar{x}_s$ versus V locus, under a sweep in the voltage level V from 0.4 V to 0.7 V in 5 mV steps. For each $(V, \bar{x}_s)$ pair, calculated numerically via (16), equation (14) (or (15)) can then be used to obtain numerically the frequency, at which the input induces an infinitesimally-small periodic oscillation in the state x about $\bar{x}_s$. Fig. 3(b) draws $f_{1‰}$ as a function of V, where $f_{1‰}$ is the input frequency, at which the steady-state time waveform of the state x features a peak-to-peak amplitude $\Delta x^{(+)} + |\Delta x^{(-)}|$ to within 1‰ of the respective mean value $\bar{x}_s$. As expected, $f_{1‰}$ increases monotonically as the input amplitude increases. The colored round symbols in Fig. 3(a) and (b) (see points $\alpha - \zeta$), respectively indicate the $\bar{x}_s$ values and the $f_{1‰}$ frequencies corresponding to zero-mean AC periodic square-wave voltage inputs with amplitudes in the set $V \in \{0.45, 0.5, 0.55, 0.6, 0.65, 0.7\}$ V. Let us now compare these results, with the ones obtained by integrating Strachan's DAE set model.

Fig. 3(c) plots simulated time responses of the TaO_x-based memristor state to square-wave purely-AC periodic voltage inputs with the same amplitude-frequency pairs as the ones indicated by the colored round symbols in Fig. 3(b). In all these simulations the device state was initialized at $x_0 = 0.25$. Depending on the input signal properties, x evolves along a different trajectory, ultimately undergoing steady-state oscillations about a distinct mean value $\bar{x}_s$. This is more clearly illustrated by the insets $\alpha - \zeta$, which zoom in across the last three periods of the device state waveforms. The dashed horizontal lines in plots $\alpha - \zeta$ reveal the mean values of the respective periodic oscillations, which are identical to the ones indicated in Fig. 3(a) up to four decimal places. Additionally, each of the steady-state periodic waveforms in these insets has a peak-to-peak amplitude, that is to within 1‰ of its mean, further validating the accuracy of the proposed equations (14) - (16) under high-frequency square-wave zero-mean periodic voltage inputs. We should note here, that since both the memristor state x and the input voltage v in the state evolution function $g(x, v)$ (see equation (4)) are practically constant over each half period of a high-frequency square-wave zero-mean periodic voltage input, the resulting asymptotic state response can be predicted to unfold along a periodic triangular envelope over time (see insets $\alpha - \zeta$ in Fig. 3(c)).

Very interestingly, the mean value $\bar{x}_s$ of the steady-state oscillation in the memristor state x for a given amplitude V of a purely-AC periodic high-frequency square-wave

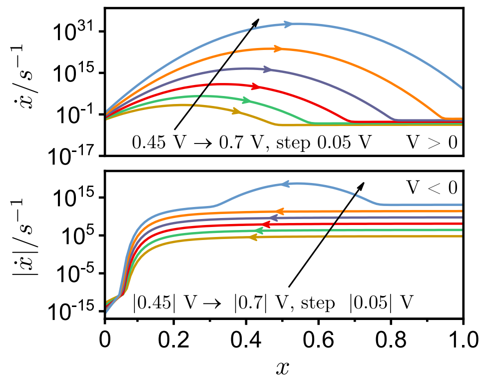

Fig. 4. DRM of the TaO_x-based memristor model under investigation. The upper (lower) panel characterizes the device on(off)-switching kinetics by depicting the SDRs corresponding to six positive (negative) input voltage levels with modulus defined in the set {0.45, 0.5, 0.55, 0.6, 0.65, 0.7} V.

voltage input can be equivalently computed by employing a slightly-modified version of the *dynamic route map* (DRM) graphical tool. Before explaining how this can be achieved, let us first provide a brief description of this powerful system-theoretic method, which has been systematically employed to study the dynamics of first-order memristors [1], [13], [20], [21], as well as to design first-order memristor-based circuits and systems [12]. For any first-order memristor, the DRM is obtained by plotting the time derivative $\dot{x}$ of the state variable as a function of the state variable x itself, for different DC input levels. It is therefore composed of a family of arrowed[4] curves, which are referred to as *state dynamic routes* (SDRs). A DRM is strictly associated with the memristor model. In particular, being influenced exclusively by the device state equation, the DRM allows to draw a comprehensive picture of the dynamical evolution of the memristor state from any initial condition and under each admissible input level. Importantly, it reveals the steady-state response of the device to any possible DC stimulus. It is worth mentioning that the first-order DRM concept was recently extended to portray the dynamics of second-order systems. The interested reader may refer to [12], [22], and [23], where the second order DRM_2 analysis tool is introduced and thoroughly explained. The DRM of the TaO_x-based memristor is displayed in Fig. 4. We notice that the device on- and off- switching kinetics, activated by positive, and negative voltage inputs, respectively, are asymmetric, in the sense that, for each $x \in [0, 1]$, in general, $g(x, V) \neq |g(x, -V)|$, which is a common feature for real-world memristors, and a crucial device characteristic for the emergence of fading memory phenomena, as already discussed in [11], and [20]. As a matter of fact, if $g(x, V) = |g(x, -V)|$, for a given zero-mean square-wave periodic voltage input amplitude V, equation (16) would be satisfied irrespective of the initial condition for the device state-variable x. In other words, x would be oscillating periodically for all $t \geq 0$, and, thus, the device would be incapable to exhibit a fading memory. It happens, though, that the asymmetry between the on- and off-switching dynamics, observed in the TaO_x nanodevice, is ubiquitous in filamentary valence change mechanism

[4]Arrows, pointing to the east (west) on the upper (lower) half plane, are pinpointed along each $\dot{x}$ versus x locus.

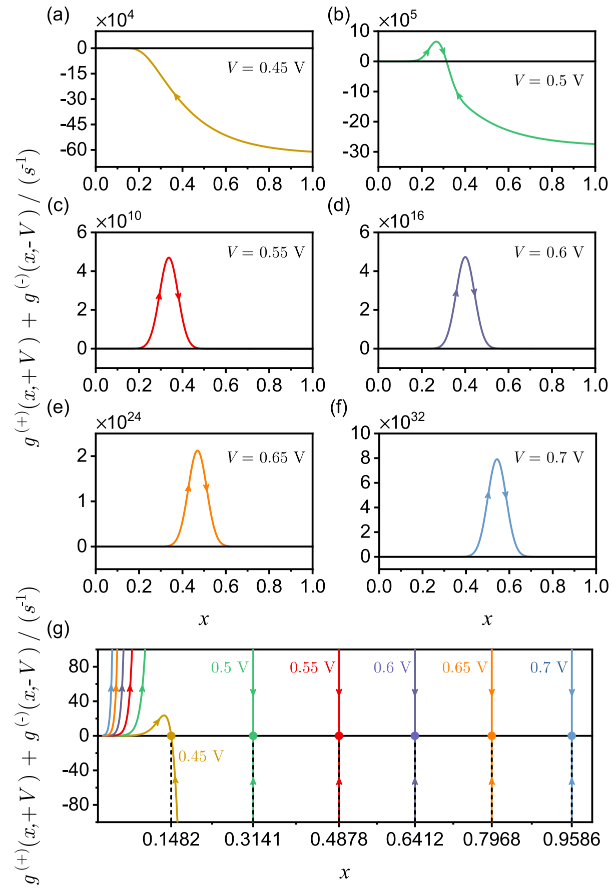

Fig. 5. Panels (a) - (f) visualize the loci of $g^{(+)}(x, V) + g^{(-)}(x, -V)$ versus the memristor state $x \in [0, 1]$ for six V-values, separated by 50 mV steps from 0.45 V to 0.7 V. Panel (g) reveals the points where the loci in panels (a) - (f) cross the horizontal axis. Each of these points shows the mean value of the steady-state oscillation in the memristor state for one of the earlier-specified amplitude levels, assigned to a square-wave purely-AC periodic high-frequency stimulus, as predicted by equations (14) - (16) (see Fig. 3(a)), and confirmed numerically (see the insets $\alpha - \zeta$ in Fig. 3(c)). The significance of the diagrams shown is elaborated in more detail in the analysis in Section III-B.

(VCM)-type memristors, which are fabricated as insulating metal oxide layers sandwiched between two metal electrodes. Switching, in this class of devices, is attributed to a voltage-induced drift-diffusion process, involving oxygen vacancies within their oxide layers. The on-switching process in VCM devices is abrupt, due to a positive feedback mechanism, whereby current increase and Joule heating reinforce each other [24]. Comparatively, the off-switching process is more gradual, due to the negative thermal feedback effect, which sets in, as the device progressively turns off, eventually resulting in an equilibrium distribution for the oxygen vacancies, when the counteractive drift and diffusion forces balance out [25]. Taking a careful look at equation (16), we notice that the mean value $\bar{x}_s$, corresponding to a given input amplitude V, is also identifiable as the point, where the $g^{(+)}(x, V) + g^{(-)}(x, -V)$ versus x locus intersects the state variable axis with a negative slope. Interestingly, this locus can be derived by adding the SDRs corresponding to input voltages V and $-V$. This clearly reveals the significance of the DRM graphic tool in this research. Panels (a), (b), (c), (d), (e), and (f) in Fig. 5 depict the $g^{(+)}(x, V) + g^{(-)}(x, -V)$ versus $x \in [0, 1]$ loci for each V-value from the set $\{0.45, 0.5, 0.55, 0.6, 0.65, 0.7\}$ V. Panel (g) in the same figure highlights the location of the intersection point of each of the loci in panels (a) - (f) with the state variable axis (see the colored round symbols). Let us now examine the predictive capability of equations (11)-(13) under sinusoidal and triangular high-frequency purely-AC periodic voltage inputs.

2) Sinusoidal and Triangular Purely-AC Periodic Voltage Inputs: We will start our analysis by pinpointing an important property of equations (11)-(13): solving this system of equations with respect to $\bar{x}_s$, the resulting solution depends on the input amplitude $\hat{v}$, as well as on the zero-mean periodic voltage input class,[5] but not on the input frequency. In other words, irrespective of the frequency value inserted in (12) and (13), the state mean value $\bar{x}_s$ corresponding to a given input amplitude $\hat{v}$, or, equivalently, the input amplitude $\hat{v}$ corresponding to a given state mean value $\bar{x}_s$, such that (11) is satisfied, will always be the same.[6] For example, the numerical solutions of (11)-(13) with respect to $\bar{x}_s$ were found to be equal to 0.1467, 0.2953, 0.4788, 0.6342, 0.7908, 0.9534, under sinusoidal purely-AC periodic voltage inputs with unitary frequency, and amplitudes defined as 0.45 V, 0.5 V, 0.55 V, 0.6 V, 0.65 V, 0.7 V, respectively. The values of the $f_{1‰}$ frequency corresponding to the aforementioned $(\hat{v}, \bar{x}_s)$ pairs were numerically calculated via (12) and found to be respectively equal to $30.74 \cdot 10^3$ Hz, $1.48 \cdot 10^8$ Hz, $1.08 \cdot 10^{10}$ Hz, $4.49 \cdot 10^{11}$ Hz, $1.74 \cdot 10^{13}$ Hz, and $6.74 \cdot 10^{14}$ Hz. In order to validate our theoretical results, we integrated the TaO_x model DAE set under sinusoidal purely-AC periodic voltage inputs with the aforementioned amplitude-$f_{1‰}$ frequency pairs, and measured the values of $\bar{x}_s$ at the end of all the resulting numerical simulations. The numbers obtained by integrating the model DAE set, and the previously reported ones, derived via equations (11)-(13), were found to be identical up to four decimal places, fully verifying the accuracy of our method.

We repeated exactly the same kind of numerical investigation, this time considering zero-mean triangular AC periodic voltage signals as inputs to the Strachan memristor model. Numerically, through equations (11)-(13), the following $(\hat{v}, f_{1‰}, \bar{x}_s)$ triplets were first predicted: (0.45 V, $5.88 \cdot 10^3$ Hz, 0.1455), (0.50 V, $2.38 \cdot 10^7$ Hz, 0.2711), (0.55 V, $2.08 \cdot 10^9$ Hz, 0.4697), (0.6 V, $8.33 \cdot 10^{10}$ Hz, 0.6272), (0.65 V, $3.11 \cdot 10^{12}$ Hz, 0.7848), (0.7 V, $1.15 \cdot 10^{14}$ Hz, 0.9481). Once again, simulating the model DAE set with the amplitude-$f_{1‰}$ frequency pairs, included in the above set of triplets, resulted in identical $\bar{x}_s$-values, up to four decimal places. What is particularly interesting is that the $\bar{x}_s$-values, calculated via (16), are quite close to the ones obtained for sinusoidal and triangular high-frequency purely-AC periodic voltage inputs via numerical simulation (see Table II).[7] Specifically, the mean absolute percentage error between the numbers, calculated via (16) and shown in column 4 from Table II, and the ones obtained by integrating the model DAE set, and shown in column 3 from the same table, is 1.93 % and 4.38 % for the sinusoidal and triangular input voltage cases, respectively. Combining columns 3 and 4 from Table II, it is clear that (16) allows to predict rather well the mean value $\bar{x}_s$ of the steady-state oscillation in the memristor state x even under high-frequency purely-AC periodic voltage stimuli of sinusoidal and triangular shape. In fact, because of the highly non-linear dependency of the device state equation on the input voltage, the memristor state exhibits noticeable changes under high-frequency purely-AC periodic inputs, only when the stimulus approaches its absolute maximum and minimum values. The multi-decade variation rate $\dot{x}$ of the memristor state x for slight changes in the input voltage v, is clearly inferable from Fig. 6, which illustrates the $\dot{x}$ versus $v \in [-0.8, 0.7]$ V loci, for $x \in \{0.2, 0.4, 0.6, 0.8, 1\}$, as calculated via equations (4)-(6). Therefore, (16), or equivalently, the modified version of the DRM, illustrated earlier in Fig. 4, can be used to provide a good estimate of the level, around which the steady-state high frequency state response of the memristor is bound to

TABLE II

MEAN VALUES OF THE STEADY-STATE OSCILLATIONS OF THE TaO_X-BASED MEMRISTOR STATE UNDER SQUARE-WAVE, SINUSOIDAL, AND TRIANGULAR HIGH-FREQUENCY ZERO-MEAN PERIODIC VOLTAGE INPUTS, AS OBTAINED BY THE STRACHAN MODEL (COLUMN 3) AS WELL AS VIA EQUATION (16) (COLUMN 4)

Input voltage amplitude $\hat{v}$	Input waveform	$\bar{x}_s$ via model simulation	$\bar{x}_s$ via eq. (16)
0.45 V	Square	0.1482	0.1482*
	Sinusoidal	0.1467	
	Triangular	0.1456	
0.5 V	Square	0.3141	0.3141*
	Sinusoidal	0.2953	
	Triangular	0.2712	
0.55 V	Square	0.4878	0.4878*
	Sinusoidal	0.4788	
	Triangular	0.4697	
0.6 V	Square	0.6412	0.6412*
	Sinusoidal	0.6342	
	Triangular	0.6272	
0.65 V	Square	0.7968	0.7968*
	Sinusoidal	0.7908	
	Triangular	0.7848	
0.7 V	Square	0.9586	0.9586*
	Sinusoidal	0.9534	
	Triangular	0.9482	
% error**	Square		0 %
	Sinusoidal		1.93 %
	Triangular		4.38 %

*Irrespective of the input waveform.
**Mean absolute percentage error between predictions, drawn from equation (16), and model numerical simulation results, for each input class.

[5]The square-wave, sinusoidal, and triangular signals define the most common classes, stimuli in electrical circuits are drawn from.

[6]Note that (16), which was simplified from (11) under the hypothesis of a square-wave high-frequency purely-AC periodic voltage input, does not include the input frequency variable either. Very importantly, however, the higher is the input frequency assigned to a general purely-AC periodic voltage stimulus, and the more accurate will be the theoretical predictions derived from (11)-(13).

[7]The high frequency assigned to the stimulus, in each curve, to derive column 3 in Table II, was set to $f_{1‰}$, as obtained via equations (12) or (13).

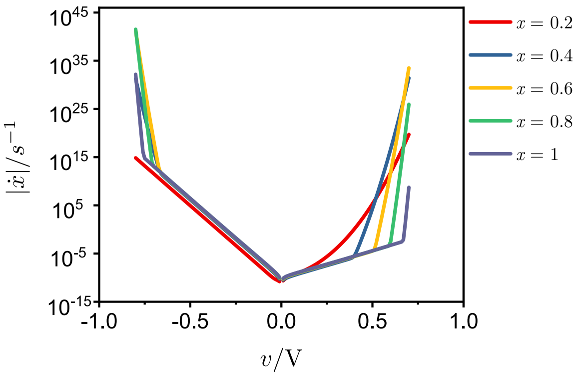

Fig. 6. Memristor switching sensitivity to the input voltage. The illustrated loci capture the dependency of the state variable time derivative $\dot{x}$ on the input voltage $\upsilon \in [-0.8, 0.7]$ V, for each x value from the set $\{0.2, 0.4, 0.6, 0.8, 1\}$. The curves were derived through the state equation (4), with $g^{(+)}(x, \upsilon)$ and $g^{(-)}(x, \upsilon)$ expressed by equations (5) and (6), respectively.

oscillate under the most common purely-AC periodic voltage stimuli used in real device experiments [13].

3) Square-Wave AC Periodic Voltage Inputs With DC Offsets: The analysis performed so far, elucidates the dynamical behavior of the TaO_x memristor under high-frequency zero-mean periodic voltage inputs. It also provides simple expressions (see (11)-(13) and (14)-(16)), that could facilitate the development of novel experimental schemes dedicated to real-world memristor state tuning. Focusing on the TaO_x nanodevice, investigated in this paper, the plots illustrated in Figs. 3(a) and (b) explicitly identify the purely-AC periodic square-wave voltage input parameters required to program this voltage-controlled device to a given state, allowing an error, due to its steady-state fluctuations, to within ± 0.5 ‰ (i.e. $\Delta x^{(+)} = |\Delta x^{(-)}| \leq 0.5‰ \cdot \bar{x}_s$, as established by the definition of the frequency $f_{1‰}$ provided in Section III-A.1). Nevertheless, the proposed system-theoretic methodology for memristor state tuning may be subject to certain limitations, depending on the device under test, with reference to the TaO_x non-volatile memristor from HP Labs. As illustrated in Fig. 3(a), the use of high-frequency purely-AC square-wave periodic voltage inputs with amplitudes in the closed range $V \in [0.4, 0.7]$ V allows to program the state within the closed subset $[0.1482, 0.9586]$, which is certainly wide, but does not cover the whole existence domain $[0, 1]$. Additionally, we notice that, for programming the state x to each $\bar{x}_s$-value within the set $\{0.5, 0.55, 0.6, 0.65, 0.7\}$, the frequency, to be assigned to the square-wave stimulus so as to reduce the error must be about and over 1 GHz (see points β, γ, δ, ϵ, and ζ in Figs. 3(a) and (b)). At these frequencies, device parasitic effects, especially from intrinsic capacitances, can be quite significant. As it will be shortly demonstrated, this issue may be resolved through the addition of a DC component to the purely-AC periodic voltage stimulus, reducing the frequency necessary to keep the error $\Delta x^{(+)} - \Delta x^{(-)}$ below $1‰ \cdot \bar{x}_s$ in each of the five aforementioned scenarios and in any other similar situation.

Assuming the voltage input is an AC periodic signal with DC offset V_o, and AC component corresponding to a square waveform with amplitude that alternates at a given frequency between two fixed minimum and maximum levels, i.e. $-V$ and V, respectively (see Fig. 7(a)), (16) is slightly modified to include the offset variable, taking the following form:

$$g^{(+)}(\bar{x}_s, V + V_o) + g^{(-)}(\bar{x}_s, -V + V_o) = 0. \tag{17}$$

Similar amendments are applied to equations (14) and (15) giving

$$\Delta x^{(+)} = g^{(+)}(\bar{x}_s, V + V_o) \cdot \frac{T}{2}, \tag{18}$$

and

$$\Delta x^{(-)} = g^{(-)}(\bar{x}_s, -V + V_o) \cdot \frac{T}{2}, \tag{19}$$

respectively. The 3D plot in Fig. 7(b) was derived by numerically solving (17) for $\bar{x}_s$, while sweeping the purely-AC periodic input amplitude V from 0.43 V to 0.48 V in 2 mV steps, and the DC offset V_o from -0.3 V to 0.3 V in 1 mV steps. *The three-dimensional (3D) illustration shows that, thanks to the inclusion of the DC offset as a new input parameter, $\bar{x}_s$ can assume any value from the admissible closed set* $[0, 1]$. Additionally, each $\bar{x}_s$ value in the state existence domain can be obtained for various combinations of input amplitude V and DC offset V_o. The colored curves, sampling the 3D surface, capture the V versus V_o dependency for each fixed $\bar{x}_s$ value within the set $\{0.1, 0.2, \ldots, 0.9\}$. These curves are also projected on the figure base, i.e. on the V_o versus V plane, so as to provide another visualization perspective. For each amplitude V of the purely-AC periodic square wave component of the voltage input signal from the set $\{0.43, 0.44, 0.45, 0.46, 0.47, 0.48\}$ V, Fig. 7(c) depicts the DC input bias voltage V_o, allowing $\bar{x}_s$ to be programmed to any value within the set $\{0.1, 0.2, \ldots, 0.9\}$, as well as the corresponding value for the input frequency $f_{1‰}$. Each frequency $f_{1‰}$ value was calculated numerically for a given triplet of $(\bar{x}_s, V, V_o)$ via equation (18) (equivalently equation (19) could have been used).

According to the information provided visually in Fig. 7(c), one way of programming the TaO_x memristor into either of the states from the set $\{0.1, 0.2, \ldots, 0.9\}$ with an amplitude percentage error $\Delta x^{(+)} - \Delta x^{(-)}$ smaller than or equal to ≤ 1 ‰ of the target state value requires: a) to choose one of the six values for V in the earlier-specified set; b) to configure the DC offset V_o according to the position of the symbol, referring to the target state value, along the V-labeled $\bar{x}_s$ versus V_o locus, and c) to assign conservatively a value to the frequency of the purely-AC periodic voltage input, which is larger than the highest $f_{1‰}$ value on the appropriate V-labeled $f_{1‰}$ versus V_o locus. Among the nine $\bar{x}_s$ values, addressed in Fig. 7(c), it is instructive to point out that $\bar{x}_s = 0.3$ is always linked to the highest $f_{1‰}$ frequency for any of the six V-levels under focus (see the colored up-ward triangles in the same figure). Following this procedure, we integrated the DAE set model of the TaO_x memristor under AC periodic square-wave voltage inputs with fixed amplitude $V = 0.45$ V, each DC offset value V_o from the set $\{-0.04283, 0.00868, 0.02016, 0.04006, 0.06625, 0.09575, 0.12673, 0.15811, 0.18929\}$ V according to the positions of the green-colored markers associated to the programming states under target (Fig. 7(c)). The input

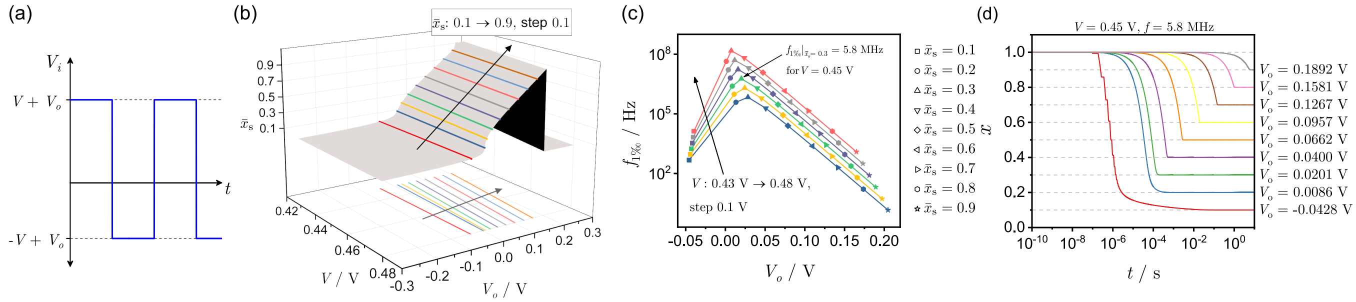

Fig. 7. Panel (a) visualizes the time waveform of an exemplary AC periodic square-wave voltage input v_i with arbitrary frequency f, amplitude V, and offset V_o. The 3D surface in panel (b) illustrates the dependence of the mean value $\bar{x}_s$ of the steady-state oscillation in the state of the TaO_x memristor under a high-frequency square-wave AC periodic voltage input V_i with non-zero mean on the amplitude V of the purely-AC input component and the DC input offset V_o. The sweep ranges for each of the input parameters V and V_o are defined in the body text. The colored curves sampling the 3D surface, capture the dependency of V on V_o for fixed $\bar{x}_s$ values, as defined in the figure legend (see also the projections of the curves on the V_o versus V plane). Panel (c) plots the frequency $f_{1‰}$ as a function of the DC offset V_o, under a given set of input amplitude levels, as specified on the lower left part of the plot, for each $\bar{x}_s$-value, identified by a particular marker, as indicated outside the plot area. The arrow on the upper right part of the plot indicates the frequency $f_{1‰}$ calculated for $\bar{x}_s = 0.3$ and $V = 0.45$ V. Panel (d) illustrates the memristor state time responses to square-wave AC periodic voltage inputs, that share the same amplitude $V = 0.45$ V and frequency $f = 5.8$ MHz, but have different DC offset V_o values, as depicted in the figure legend. In all cases, the device state was initialized at the lowest possible resistive state, that is when $x(0) \triangleq x_0 = 1$.

frequency was set to $f_{1‰}|_{V=0.45\,\mathrm{V},\,\bar{x}_s=0.3}$, i.e. to 5.8 MHz (see the peak of the $f_{1‰}$ versus V_o locus for $V = 0.45$ V in the same figure). Having initialized the device in all cases at its highest conduction state ($x_0 = 1$), the resulting time waveforms of the memristor state are illustrated in Fig. 7(d). The results fully verify the system theory-assisted analysis methodology presented throughout this section. Importantly, adding a DC offset to the zero-mean AC periodic square-wave voltage input, we were able to reduce significantly the input frequency required to program the TaO_x memristor to any state within the $[0, 1]$ existence domain under the accuracy requirements established through the definition of the frequency $f_{1‰}$. The input frequency could be further reduced by assigning a smaller value to the amplitude V of the purely-AC periodic component of the input voltage signal, e.g. 0.43 V or 0.44 V, as indicated in Fig. 7(c). As a final remark, we pinpoint that the methodology presented in this section can be more generally applied to any AC periodic input voltage with a DC offset, including sinusoidal and triangular waveforms.

B. Transient Response

Revisiting the simulation experiments performed in Section II-B, Fig. 8(a)((b)) zooms in on the time waveform from Fig. 2(a) for $x_0 = 0.2$ ($x_0 = 1$) over the interval $0 \leq t \leq 0.04\,\mu$s, to depict in detail the increase $\Delta x^{(+)}$, decrease $\Delta x^{(-)}$, and net change Δx, in the memristor state, as well as its mean value $\bar{x}$, over each of the first four cycles of the voltage stimulus $v = (0.5\,\mathrm{V})\sin(2\pi(100\,\mathrm{MHz})t)$ (see panel (c)). We have already verified the assumption by which after transients decay to zero, the state variable x is practically equal to the mean value $\bar{x}_s$ of its steady-state oscillation, under a high-frequency periodic voltage input. By taking a careful look at the blue and orange x vs t loci illustrated in Figs. 8(a) and (b), respectively, we notice that, in both, the maximum swing of the state x (i.e. $\Delta x^{(+)}$, $|\Delta x^{(-)}|$) over each of the depicted cycles, is much smaller than its mean value $\bar{x}$ over the same cycles, which is indicated through a dashed horizontal line. It is therefore reasonable to assume also that the state variable x is approximately equal to the mean value $\bar{x}_n$ of its oscillation over the n^{th} cycle of the high-frequency purely-AC periodic voltage input signal v, i.e. $x \approx \bar{x}_n$, for $t \in [(n-1)T, (n-1)T + T]$. The values of $\Delta x^{(+)}$, $|\Delta x^{(-)}|$, and $|\Delta x|$, over 2000 input cycles, for $x_0 = 0.2$ ($x_0 = 1$), are plotted in panel (d) ((e)). We notice that for $\bar{x}_n < (>) \bar{x}_s = 0.2953$ (or $n < 1758$ (1629)), the state variable is more sensitive to increases (decreases) under the positive (negative) excursion of the purely-AC periodic voltage stimulus than to decreases (increases) under the negative (positive) one, over each cycle. This nonlinear phenomenon is a direct consequence of the asymmetry between the device on- and off- switching kinetics, which is clearly visualized in the DRM from Fig. 4. Summing up our observations, we may claim that the net change Δx in the device state x, over each cycle of the purely-AC periodic high-frequency voltage stimulus $v(t)$ under focus, is linked to the running mean value $\bar{x}$ of its oscillating waveform, over the same cycle, through the following relationship:

$$\Delta x = \Delta x^{(+)} + \Delta x^{(-)}, \tag{20}$$

where

$$\Delta x^{(+)} = \int_0^{T/2} g^{(+)}\big(\bar{x}, v(t)\big)\, dt, \tag{21}$$

and

$$\Delta x^{(-)} = \int_{T/2}^{T} g^{(-)}\big(\bar{x}, v(t)\big)\, dt. \tag{22}$$

As already discussed in Section III-A, when the state variable time response $x(t)$ becomes periodic, the relationship between $\Delta x^{(+)}$ and $\Delta x^{(-)}$ is given by: $\Delta x^{(+)} = -\Delta x^{(-)}$, i.e. $\bar{x} = \bar{x}_s$, and the system of equations (20)-(22) reduces to the one expressed by equations (11)-(13).

Let us now introduce a new system-theoretic tool, namely the input-referred *High-Frequency Dynamic Route Map*

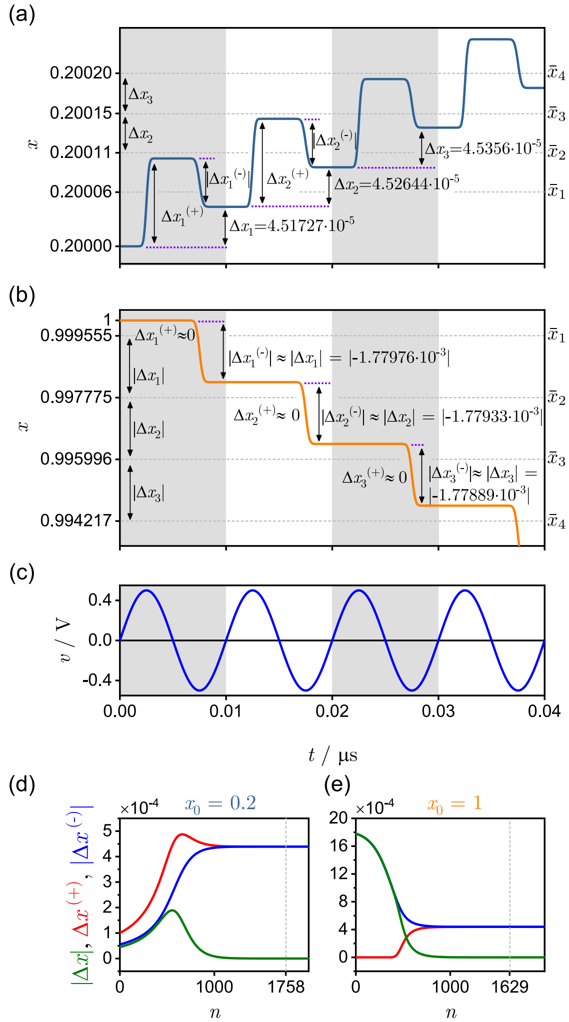

Fig. 8. Panel (a) ((b)) illustrates the first four periods in the time response of the memristor state variable $x(t)$, starting from $x_0 = 0.2$ ($x_0 = 1$), to a zero-mean sinusoidal voltage input with amplitude $\hat{v} = 0.5\,\text{V}$ and frequency $f = 100\,\text{MHz}$ (see panel (c)). $\Delta x_n^{(+)}$ and $\Delta x_n^{(-)}$, respectively represent the change in the state x due to the positive and negative excursion of the stimulus over the first and second half period of the n^{th} cycle. $\Delta x_n^{(+)}$ ($\Delta x_n^{(-)}$), also indicate the maximum swing of the state over the n^{th} period of the stimulus, for $x_0 = 0.2$ ($x_0 = 1$). Δx_n denotes the change in the memristor state over the n^{th} input cycle. The horizontal dashed lines in panel (a) ((b)) indicate the mean values of $x(t)$ over the first four input cycles, for $x_0 = 0.2$ ($x_0 = 1$). Panel (d) ((e)) plots $|\Delta x|$, $\Delta x^{(+)}$, and $|\Delta x^{(-)}|$ versus the input cycle number n for $x_0 = 0.2$ ($x_0 = 1$). The vertical dashed line indicates the cycle number, at which the state variable waveform becomes periodic.

(HF-DRM), by which the temporal evolution of the memristor state, from any initial condition, under any high-frequency periodic input, may be anticipated very accurately. The HF-DRM is determined exclusively through equations (20)-(22), and thus its form depends on the device state equation (see (4)-(6)), as well as on the shape of the periodic input $v(t)$. It is derived by plotting the set of $\Delta x/T$-values versus the respective $\bar{x}$-values, falling in the allowable state existence domain $[0, 1]$ for each admissible amplitude $\hat{v}$ of a predefined high-frequency AC periodic voltage stimulus v. Since Δx is normalized by T, the HF-DRM is frequency independent. Nevertheless, its predictions should be considered valid only when the input can be characterized as a high-frequency periodic signal. Let us now compose the HF-DRM of the TaO_x memristor, under a zero-mean high-frequency square-wave voltage input. Substituting the input amplitude $\hat{v} = V$ in the system of equations (20)-(22), results in the following expression for the normalized net change $\Delta x/T$ in the memristor state:

$$\frac{\Delta x}{T} = \frac{1}{2}\Big(g^{(-)}(\bar{x}, +V) + g^{(-)}(\bar{x}, -V)\Big). \tag{23}$$

Interestingly, the square-wave-referred HF-DRM of the TaO_x memristor, may be directly obtained from the plots illustrated in Fig. 5, by inserting $x \rightarrow \bar{x}$ and scaling down the y-axes by 2.

The HF-DRM of the TaO_x-based nanodevice, under a zero-mean high-frequency sinusoidal voltage input, is plotted in the upper panel of Fig. 9(a), for each $\hat{v}$ value, included in the set $\{0.45, 0.5, 0.55, 0.6, 0.65, 0.7\}\,\text{V}$. The lower panel in (a) provides a detailed view of the $\Delta x/T$ versus $\bar{x}$ loci in a region around the horizontal axis. Similarly to the DRM (see Fig. 4 in Section III-A.1), each curve in Fig. 9(a) is referred to as a *High-Frequency State Dynamic Route (HF-SDR)*, being associated to a particular amplitude $\hat{v}$ of the specific sinusoidal voltage stimulus under focus. Clearly, the mean value $\bar{x}_s$ of the steady-state oscillation in the state variable x, for a given input amplitude $\hat{v}$, is identified as the point where the corresponding HF-SDR crosses the horizontal axis with a negative slope (see any colored round point sitting on the $\bar{x}$ axis in the lower panel of Fig. 9(a)).

We will now show how to predict the mean value $\bar{x}$ of the state variable x, preliminarily initiated in each of the two x_0-values from the set $\{0.2, 1\}$, over 2000 cycles of the periodic input illustrated in Fig. 8(c). This will be carried out by studying the respective HF-SDR corresponding to an input amplitude $\hat{v}$ of 0.5 V (see black-shaded trace Fig. 9(a)). The aforementioned curve is re-plotted in Fig. 9(b), and partly in panels (c) and (d) of the same figure, in order to illustrate in detail the transient evolution of the value $\bar{x}$ of the oscillation in the memristor state x toward the programming target value for each of the two scenarios under focus. Visualization tools of this kind allow to gain a deeper insight into the complex dynamics of non-volatile resistance switching memories. By multiplying $\Delta x/T$ by the input period T of a high-frequency voltage stimulus of pre-defined shape, the resulting Δx versus $\bar{x}$ locus, corresponding to a specific input amplitude $\hat{v}$, now depends on the frequency of the stimulus. For example, the y-axis in each of panels (b)-(d) of Fig. 9 was scaled to represent Δx for a zero-mean periodic sinusoidal voltage input oscillating at 100 MHz. The blue (orange)-colored locus in the same panels shows the transient evolution of the point trajectory ($\bar{x}$,Δx) under the application of the high-frequency purely-AC periodic sinusoidal voltage stimulus shown in Fig. 8(c). The TaO_x memristor was preliminarily programmed into a state x_0-value of 0.2 (1). The blue (orange)-colored locus in panels (b)-(d) was extracted through an alternative algorithm, as proposed next.

The mean value $\bar{x}$, of the time waveform of the state variable x over the first cycle ($n = 1$) of the stimulus of Fig. 8(c) is focused to be equal to 0.200062 (0.999555) for $x_0 = 0.2$

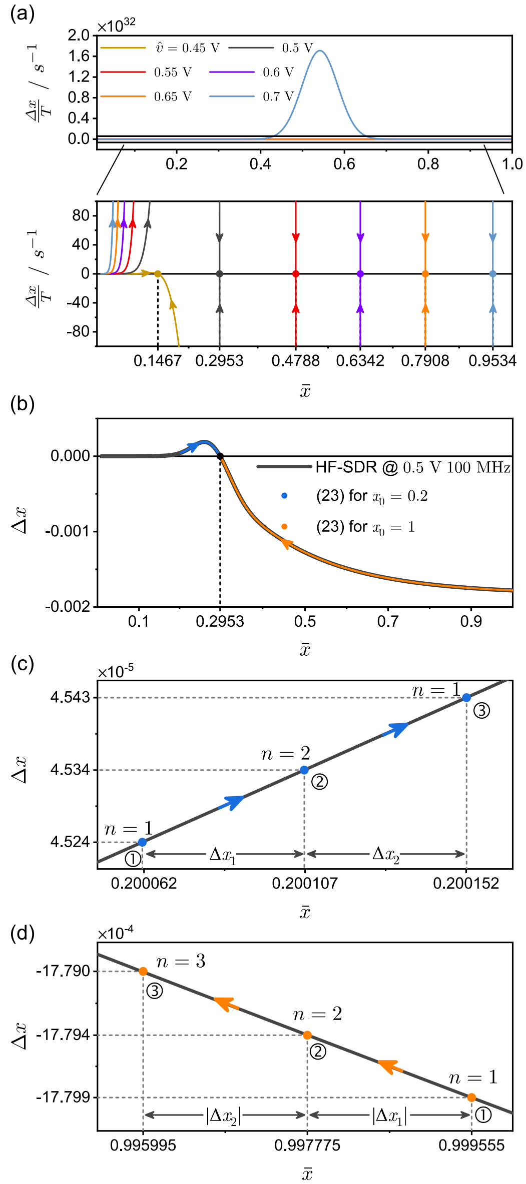

Fig. 9. (a) Upper panel: HF-DRM of the TaO_x-based memristor, associated to a high-frequency purely-AC sinusoidal periodic voltage stimulus υ, for each input amplitude $\hat{\upsilon}$ in the set $\{0.45, 0.5, 0.55, 0.6, 0.65, 0.7\}$ V. This HF-DRM was derived through the system of equations (20)-(22). (a) Lower panel: zoomed in view of the HF-DRM, shown in the upper panel, about the horizontal axis. Colored round symbols depict the points where the HF-SDRs cross the horizontal axis with negative slopes. Panel (b) shows, through a black-shaded arrowed trace, a frequency-scaled variant of the HF-SDR, extracted from the HF-DRM of Fig. 9(a) for an input amplitude $\hat{\upsilon}$ of 0.5 V, and adapted to an input period T of 10 ns. This input frequency-dependent locus predicts how the change Δx in the memristive state per input period depends on the mean value $\bar{x}$ of the time waveform of the memristor state itself per input period, for all the 2000 cycles of the 100 MHz-sine wave voltage stimulus from Fig. 8(c). The blue (orange) arrowed locus, found to lie over part of the frequency-scaled HF-SDR under focus, was derived through the proposed iterative algorithm, mathematically described by equation (24) (refer to the body text from more details). Panel (c) ((d)) illustrates the transient evolution of the trajectory point $(\Delta x, \bar{x})$ over the first three iterations of the proposed algorithm for $x_0 = 0.2$ ($x_0 = 1$). The arrows along the blue (orange) locus in panel (c) ((d)), point toward the east (west) indicating a positive (negative) net change Δx_1 in the memristor state x over each cycle i of the voltage stimulus in Fig. 8(c) ($i \in \{1, 2, 3\}$) for $x_0 = 0.2$ ($x_0 = 1$).

($x_0 = 1$), as follows from the numerical integration of the TaO_x model equations, which resulted in the plot in Fig. 8(a) (8(b)). Together with the knowledge of the change Δx_1 in the state x, over the first input cycle, which may be deduced via (21)-(22), the earlier determined mean value $\bar{x}$, of the time waveform of the state x over the first input cycle allows to identify a first point on the Δx versus x plane. In regard to the 0.5 V 100 MHz sine wave input case study under focus, this procedure allows to identify the first point ① on the blue (orange) locus in Fig. 9(b), as marked in the zoom in view of Fig. 9(c) (9(d)) for $x_0 = 0.2$ ($x_0 = 1$). Based on an earlier assumption, according to which the state may be approximated by its mean value over each cycle of the high-frequency stimulus for either $x_0 = 0.2$ or $x_0 = 1$, the mean value $\bar{x}_2$ of the state over the second cycle ($n = 2$) of the stimulus in Fig. 8(c) may be predicted by adding the ordinate Δx_1 of point ① to its abscissa $\bar{x}_1$, while the change Δx_2 in the state x over the second input cycle may be deduced via (21)-(22). As shown in Fig.9(c) (9(d)), for the simulation scenario from Fig. 8(a) (8(b)), a second blue (orange) point ②, specifically $(\bar{x}_2, \Delta x_2)$, may be marked on the same plane, where the sign-referred HF-SDR for 0.5 V 100 MHz, is visualized. For each of the two case studies, the mean value $\bar{x}_3$ of the state x over the third cycle ($n = 3$) of the stimulus of Fig. 8(c) may then be predicted by summing the ordinate Δx_2 of point ② to its abscissa $\bar{x}_2$, while the change Δx_3 in the state x over the third input cycle may be deduced via (21)-(22). As indicated in Fig. 9(c) (9(d)), for the simulation scenario from Fig. 8(a) (8(b)), a third blue (orange) point ③, specifically $(\bar{x}_3, \Delta x_3)$, may then be marked in Fig. 9(c) (9(d)). The procedure may then be continued similarly. Impressively, in Fig. 9(c) (9(d)), the set of blue (orange) points $\{(x_i, \Delta x_i), i \in N > 0\}$ is found to lie along the sign-referred HF-SDR for 0.5 V 100 MHz, validating the proposed iterative algorithm, which can be mathematically described by the following discreet map:[8]

$$\bar{x}_{n+1} = \bar{x}_n + \Delta x_n, \tag{24}$$

where Δx_n is calculated in each step numerically through (11)-(13).[9] As this iterative process continues, $\bar{x}$ and Δx will converge asymptotically to $\bar{x}_s$ and zero, respectively. This is clearly inferable from the blue and orange loci in Fig. 9(b), which were separately derived by iterating the proposed algorithm for 2000 input cycles with either of the two earlier specified initial conditions for the memristor state. Note that the values for Δx_n and $\bar{x}_n$, computed numerically through equation (24), for the first 3 input cycles ($n = 3$) for $x_0 = 0.2$ ($x_0 = 1$) - refer to panel (c) ((d)) in Fig. 9 - are almost identical to those calculated from the model numerical simulation results indicated in Fig. 8(a) (8(b)). Very interestingly, even though equations (21)-(22) assume that the state variable x keeps constant, and equal to $\bar{x}_n$ over each input cycle n, the error in the estimate of the net change Δx

[8]Very importantly, loci of the kind shown in Fig. 9(c) or (d), and derived through an iterative algorithm, such as the one, based on (24), is reminiscent of the cobweb plot typically used in the theory of nonlinear dynamics [26] to study discrete maps.

[9]Notice that, having plotted the sign-referred HF-SDR for 0.5 V 100 MHz (see black-shaded arrowed traces in Fig. 9(b)-(d)), it is possible to implement the proposed iterative procedure solely graphically, since any mean value $\bar{x}_n$ determines a corresponding Δx_n value, and thus, a specific point on the aforementioned curve.

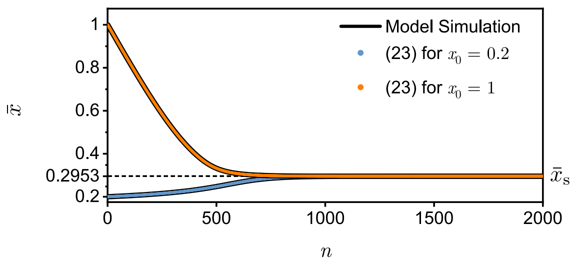

Fig. 10. Running mean value $\bar{x}$ of the oscillation in the memristor state x over 2000 cycles of the high-frequency purely-AC periodic sine wave voltage stimulus of Fig. 8(c) for each initial condition x_0 from the set {0.2, 1} as observed in model numerical simulations (black trace) and obtained through the discreet map of (24) (blue and orange traces, for $x_0 = 0.2$ and $x_0 = 1$, respectively).

in each iteration of the algorithm is almost negligible. For $x_0 = 0.2$ ($x_0 = 1$), the mean percentage error in Δx over 2000 iterations is only about 0.009% (0.016%), relative to the simulation result in Fig. 8(a) (8(b)). The maximum error in Δx on the other hand, is as low as 0.052% (0.094%) in the first (latter) case. The mean value $\bar{x}$ of the time waveform of the state x, over 2000 cycles of the 0.5 V 100 MHz sine wave stimulus from Fig. 8(c) ($n = 2000$), for each initial condition x_0 from the set {0.2, 1}, is shown in Fig. 10, as resulting from Strachan's model numerical integration (black traces) and from the application of the discrete map of eq. (24) (blue and orange traces for $x_0 = 0.2$ and for $x_0 = 1$, respectively). This figure clearly provides evidence for the accuracy of the theoretical analysis reported in this section.

IV. Discussion and Conclusion

The input-induced response of a real-world sign-invariant non-volatile memristor is the result of a highly-nonlinear dynamical process depending on the device switching sensitivity to input level and polarity, and to the memristor state. Together with other works available in the literature [26], this paper demonstrates how an in-depth system-theoretic investigation of a nonlinear system, such as the memristor, enables to draw a comprehensive picture for its response to input/initial condition combinations of interest to the application at hand. Specifically, this paper gains a deep understanding of the mechanisms underlying the resistance switching phenomena, occurring in a TaO_x memristor under high-frequency AC periodic stimulation. Under stimuli of this kind, the memristor state is found to periodically oscillate with a negligible peak-to-peak amplitude at steady-state. Then, during the transient phase, the memristor state undergoes a very small net change over each cycle of a high-frequency excitation signal, which simplifies the theoretical analysis of the device mathematical model, enabling its analytical integration. The theoretical analysis, performed here, allowed us to derive analytical formulas that linked very accurately the mean value $\bar{x}_s$ of the steady-state oscillation in the memristor state x from Strachan's TaO_x memristor model to the high-frequency periodic input properties, only. Specifically, we proved that under a given AC periodic testing voltage signal v, the mean value $\bar{x}_s$ of the steady-state periodic oscillation in the memristor state x depends only on the input amplitude V and offset V_o, but not on the initial state x_0, preliminarily programmed into the device, providing for the first time a rigorous mathematical proof for the fading memory effects in real-world memristors [11], [13]. This is a significant finding, since it may allow the derivation of Volterra series representations for non-volatile memristors. Such representations may lead to the development of a systematic approach to memristor device modelling and the related circuit and system design and optimization. The aforementioned analytical formulas, derived here, were then validated through a series of simulation experiments involving periodic inputs of square-wave, sinusoidal, and triangular shape. Next, we exploited our theoretical findings to develop a method for programming the TaO_x memristor to any target state from a given set by properly configuring the DC offset level V_o of a fixed V-amplitude high-frequency square-wave AC periodic voltage input. Finally, we introduced a new system-theoretic graphical tool for the analysis of first-order memristive systems driven by high-frequency AC periodic inputs, namely the input-referred *High-Frequency Dynamic Route Map*. This tool allows to determine the running mean value $\bar{x}$ of the oscillating waveform of the memristor state across each cycle of a high-frequency AC periodic voltage stimulus. It's important to note once more that the methods presented in this paper are not limited to the specific TaO_x memristor model, but may be applied to any first-order sign-invariant non-volatile memristor differential algebraic equation set, in which the state evolution function features distinct expressions. Furthermore, although the methods presented in this paper, are based on purely-theoretical results, their experimental validation is one of the goals of our future research investigations. In fact, the theoretical findings, reported in this paper, are expected to inspire new algorithms and/or techniques for programming non-volatile memristors. The analytical investigation of the high-frequency response of volatile memristors will be introduced in a follow-up publication. As recently revealed, some physical realisations of volatile memristors, allow to model high-order biological phenomena through low-order bio-inspired circuits [27], [28].

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